to one another by lymph vessels. **Lymph** is a clear, transparent, colourless, mobile and extracellular fluid connective tissue. As the lymph percolates through the lymph node, the particulate antigen brought in by the lymph will be **trapped** by the phagocytic cells, follicular and interdigitating dendritic cells.

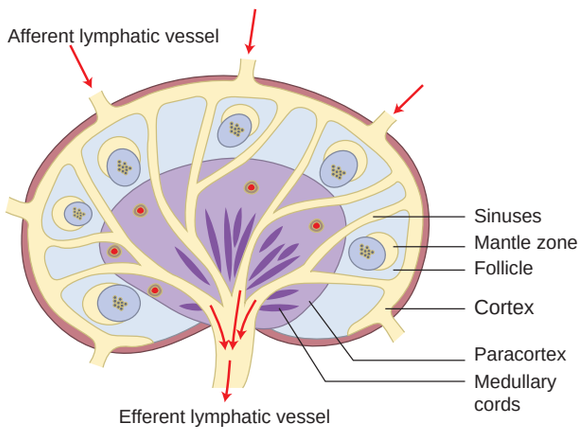


Fig. 7.13 Secondary lymphoid organ – Structure of lymph node

Lymph node has three zones **(Fig. 7.13).** They are the **cortex, paracortex and medulla**. The outer most layer of the lymph node is called cortex, which consists of B-lymphocytes, macrophages, and follicular dendritic cells. The **paracortex** zone is beneath the cortex, which is richly populated by T lymphocytes and interdigitating dendritic cells. The inner most zone is called the medulla which is sparsely populated by lymphocytes, but many of them are plasma cells, which actively secrete antibody molecules. As the lymph enters, it slowly percolates through the cortex, paracortex and medulla, giving sufficient chance for the phagocytic cells and dendritic cells to trap the antigen brought by the lymph. The lymph leaving a node carries enriched antibodies secreted by the medullary plasma cells against the antigens that enter the lymph node. Sometimes visible swelling of lymph nodes occurs due to active immune response and increased concentration of lymphocytes. Thus swollen lymph nodes may signal an infection. There are several groups of lymph nodes. The most frequently enlarged lymph nodes are found in the neck, under the chin, in the armpits and in the groin.

The mucosa-associated lymphoid tissue **(MALT)** is a diffuse system of small concentrations of lymphoid tissue in the alimentary, respiratory and urino-genital tracts. **MALT** is populated by lymphocytes such as T and B cells, as well as plasma cells and macrophages, each of which is well situated to encounter antigens passing through the mucosal epithelium. It also possesses IgA antibodies.

Gut-associated lymphoid tissue **(GALT)** is a component of the mucosa-associated lymphoid tissue **(MALT)** which works in the immune system to protect the body from invasion in the gut.

Bronchus Associated Lymphoid Tissues **(BALT)** also a component of MALT is made of lymphoid tissue (tonsils, lymph nodes, lymph follicles) is found in the respiratory mucosae from the nasal cavities to the lungs.

Cells of the immune system

The immune system is composed of many interdependent cells that protect the body from microbial infections and the growth of tumour cells. The cellular composition of adult human blood is given in **Table 7.7.**

All these cells are derived from pluripotent haematopoetic stem cells. Each stem cell has the capacity to produce RBC, WBC and platelets. The only cells capable of specifically recognising and producing an immune response are the lymphocytes. The other types of white blood cells play an important role in non specific immune response, antigen presentation and cytokine production.

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